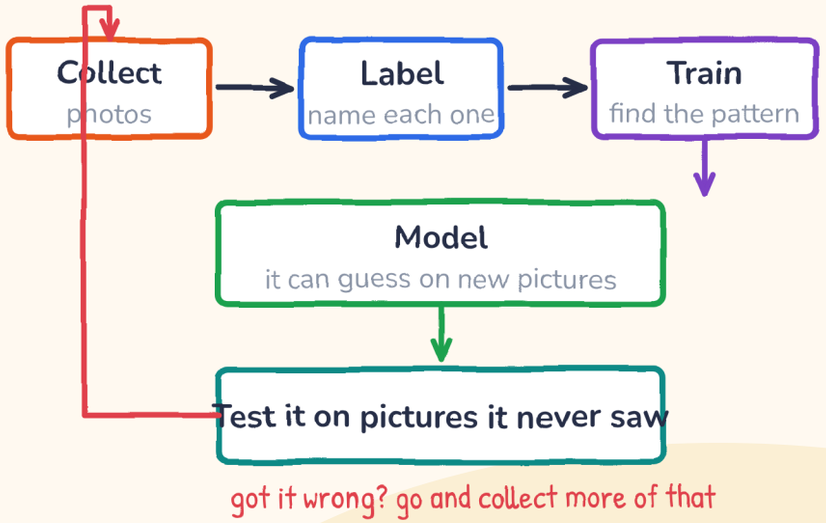
Artificial Intelligence · Machine Learning Foundations

Training Data

Teach a computer to see. It has never met a banana, and in ten minutes it will spot one.

60 minutes · Pairs · Laptop + webcam



Did you know?

In 2012 a model called AlexNet learned to recognise a thousand kinds of object from 1.2 million labelled photos, and beat every hand-written program by so far that the whole field changed direction within a year.

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Let's think

Nobody gave you rules for a banana



How did you learn it?

You can spot a banana in a second. Green, black-spotted, peeled, sliced, in a photo, in a cartoon.

Nobody ever sat you down with a list of banana rules. You just saw **a lot of bananas**, and somebody told you the word.



That is exactly the trick

Last lesson we found that rules break. Today we do the other thing: collect examples, attach the right answer to each one, and let the computer find the pattern.

Guess first

- How many photos would you need to teach it **banana**?
- Would 100 photos of the **same** banana be enough?
- What happens when it sees a banana in a dark room?
- How would you ever know if it was any good?

Write your four guesses down

We test every one of them before the end of the lesson. You will be right about some and wrong about at least one.

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Learning aims

Today I will learn to...

1 Collect and **label** training examples

2 **Train** an image model and test it on new pictures

3 Explain why **varied** examples beat more of the same

4 Measure **accuracy** honestly, on data the model never saw

Keywords

training data · label · class · sample · model · prediction · confidence · test set · accuracy · bias

Objective 1 · The big idea

Four words you need today

Class

One of the answers the model is allowed to give. **banana**, **apple**, **nothing**. You decide the classes before you start.

Label

The right answer attached to a sample. In Teachable Machine the label is simply **which box you put the photo in**.

Sample

One example. One photo. Teachable Machine counts these for you, and the count is the first thing to look at when something goes wrong.

Model

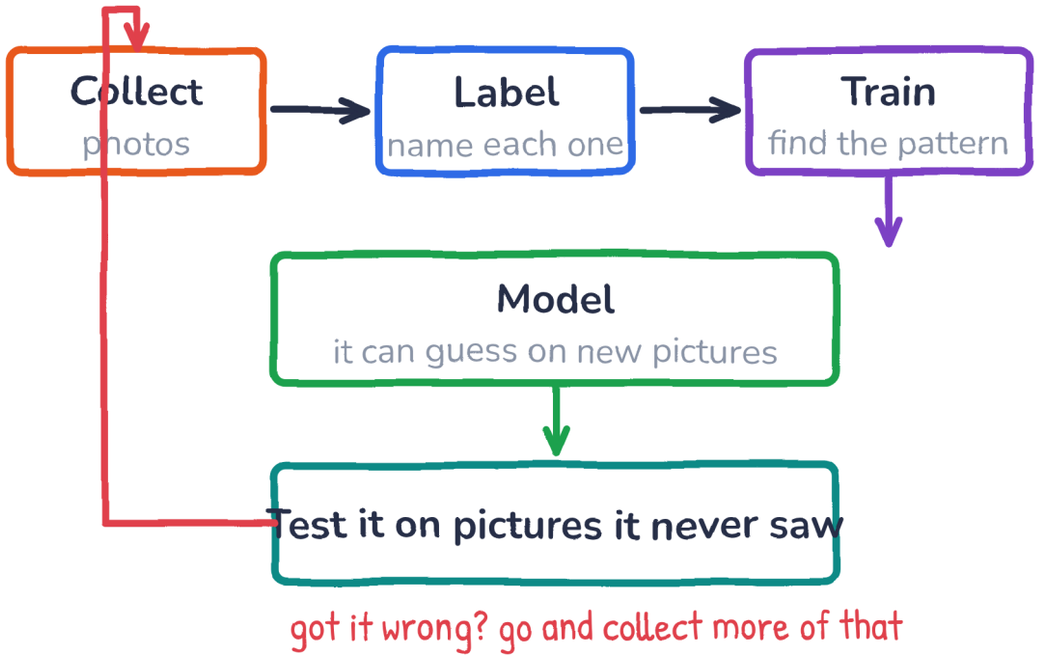
What comes out of training. A pattern, frozen, that turns a new photo into a guess and a number saying how sure it is.

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Objective 1 · The cycle

The loop every AI team runs

Collect, label, train, test, repeat



It is a loop, not a line

No team gets it right first time. They train, find what it gets wrong, go and collect **more of that**, and train again.

The loop is where the work is

Training takes seconds. Collecting the right examples takes days. Real AI jobs are mostly about data, not about maths.

Today

You will go round this loop at least twice. The second time round is where the learning happens.

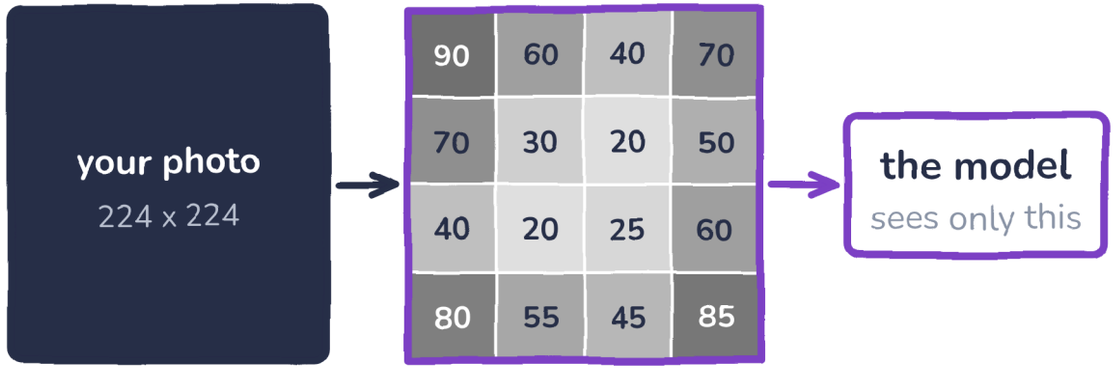
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Objective 2 · Inside

What the computer actually sees

No picture, just numbers

a picture is just a grid of numbers



50,176 pixels, three numbers each

150,528 numbers, for one small photo

A photo is a grid

Every pixel is three numbers: how much red, green and blue. A small 224 by 224 photo is **150,528 numbers**.

The pattern is in the numbers

The model never sees yellow or curved. It finds that *these* numbers tend to go together when you said banana.

Why that matters

Change the light and every number changes. That is why a model trained in one room can fail in the next one.

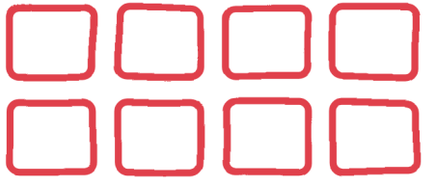
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Objective 3 · The key idea

Variety beats quantity

Twenty photos, collected two ways

20 photos, all the same

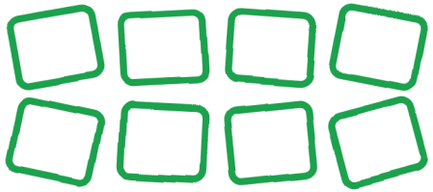


same hand, same desk, same light

New photo, new room

51% · no idea

20 photos, all different



different hands, angles, backgrounds

New photo, new room

94% · banana

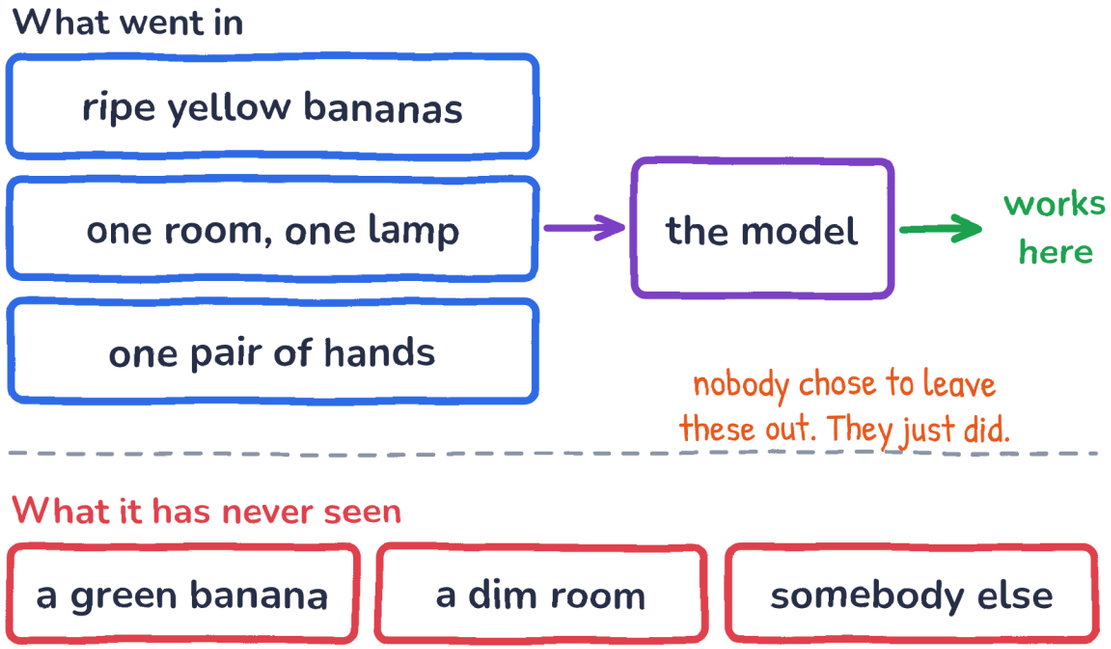
same effort, very different model

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Objective 3 · Why

A model only knows what it was shown

What went in decides what works



This has a name

When the training data leaves something out, the model is **biased** against it. Not because anyone meant it. Because nobody photographed it.

It matters outside class

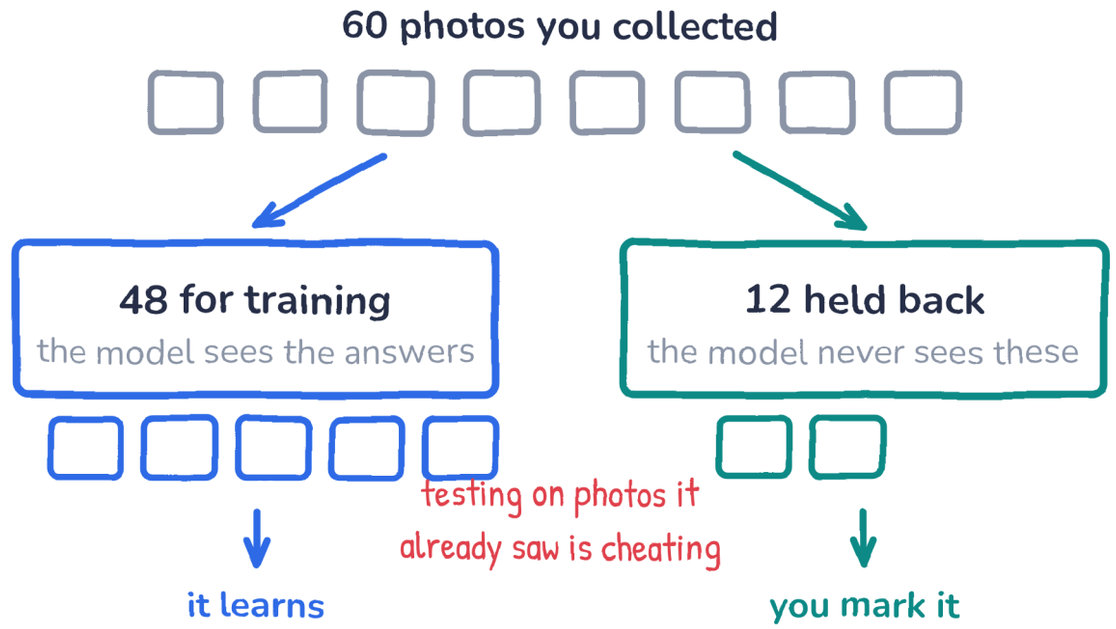
Face systems that worked on some skin tones and not others, and voice systems that missed some accents, were built by people who did not notice what their data was missing.

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Objective 4 · Honesty

Never mark your own homework

Hold some examples back



Train set and test set

Most of the photos teach the model. A handful are hidden from it and used only at the end, to find out what it really learned.

Why hold any back?

A model can memorise. Ask it about a photo it already studied and it looks perfect. The held-back photos are the only honest question you can ask.

In Teachable Machine

There is no split button. You make the test set yourself: take a few photos, keep them out, and try them at the end.

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In the real world

Who is collecting data right now

Your phone gallery

Search "mango" in your photos and it finds them. Somebody labelled millions of pictures so yours could be searched.

A mango farm

Sorting machines grade fruit by photo. They are trained on that farm's own fruit, because another farm's mangoes look different.

Every CAPTCHA

"Select all squares with a bus." You are labelling training data, free, a few seconds at a time.

Part 2 · Make it

Now teach one yourself

Three classes, a honest test, and one round of making it better.

35 minutes | Pairs | Teachable Machine



MAKE

Class game

Be the model

How to play

1. One student leaves the room. They are the model.
2. The class picks a secret class, say **things that roll**.
3. They come back. You show them objects and say only **yes** or **no**. No explaining.
4. After ten objects, hand them a new one. Can they guess?

They just learned from labelled examples, with no rules given.

Round two: bad data

Now only show them **round red** things that roll. Then hand them a blue pen lid.

They will say no. They learned *red and round*, because that is all you showed.

Say it out loud

"The model is not stupid. It learned exactly what we taught it." That sentence is the whole lesson.

MAKE

Before you start

What you need



Kit, per pair

- One laptop with a webcam
- Chrome or Edge, and internet
- **teachablemachine.withgoogle.com**
- Three objects: say a banana, an apple, and nothing at all

Two rules before you touch it

- Nothing leaves the laptop. Training runs **in the browser**.
- No photos of faces today, yours or anyone else's. Objects and hands only.

Why the third class?

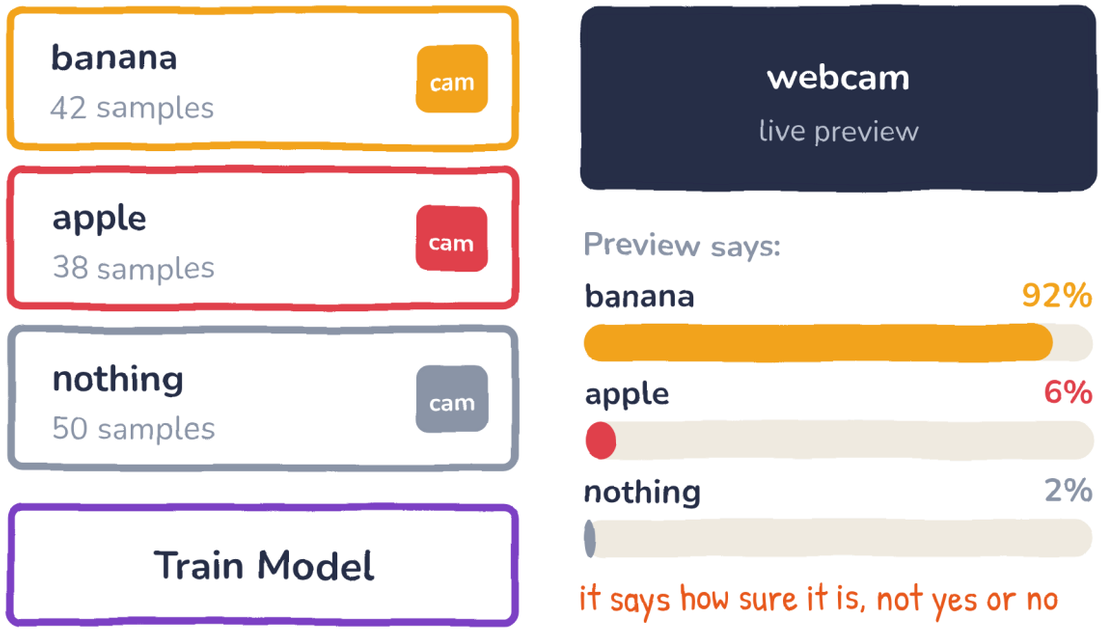
A model with only banana and apple must answer one of the two, even for an empty desk. A nothing class gives it somewhere honest to put that.

MAKE

Build step 1 of 4

Collect and label

The screen you will see



it says how sure it is, not yes or no

Do this

1. New Project, then **Image Project**, then Standard.
2. Rename Class 1 to **banana**, Class 2 to **apple**.
3. Add a Class and call it **nothing**.
4. Hold Webcam and record about **40 samples** each.
5. Keep the three counts roughly equal.

Move while you record

Turn it, tilt it, move nearer, change hands, step sideways. Forty varied samples beat four hundred identical ones.

MAKE

Build step 2 of 4

Train it, then try to break it

Do this

1. Press **Train Model**. Do not switch tabs while it runs.
2. Watch the Preview bars as you hold things up.
3. Now try to fool it: move to a window, use your other hand, hold the fruit further away.
4. Write down **three things that fooled it**.

Read the bars, not the winner

92% banana is a confident model. 48% banana and 44% apple is a model guessing. Both show the same top answer.

It never says yes or no

Every prediction is a set of percentages that add up to 100. The app just shows you the biggest one.

The honest question

Not "does it work?" but "when does it stop working?" Engineers get paid for the second question.

MAKE

Build step 3 of 4

Measure it properly

Twelve honest questions

the 12 photos it had never seen

✓ ✓ ✓ ✓ ✓ ✓

✓ ✓ ✓ ✓ ✗ ✗

10 right out of 12

accuracy = 10 ÷ 12 = 83%

write it down before you change anything

Do this

1. Take **12 new photos**: 4 banana, 4 apple, 4 empty. New spot, new light.
2. Show each one and write down what it said.
3. Count the right ones. Divide by 12.
4. That number is your **accuracy**. Write it on the board.

Compare across the class

Every pair will get a different number from the same app and the same fruit. The difference is entirely in the data they collected.

MAKE

Build step 4 of 4

Round two: make it better

Do this

1. Look at your three failures from step 2.
2. Collect **15 more samples of exactly those cases**.
3. Train again.
4. Run the **same 12 test photos**. Did accuracy go up?

This is the job

You did not change the model, the maths or the code. You changed the data. That is how nearly every real AI system gets better.

If it got worse

That happens, and it is worth more than a win. Usually one class now has far more samples than the others, so the model leans towards it.

Keep both numbers

Before and after, on the same 12 photos. A measurement you can compare is worth more than one that sounds good.

Test like an engineer

Break it on purpose

Try this	What usually happens	What it teaches
Hold the fruit against a window	Confidence collapses	Backlight changes every pixel number
Ask a different student to hold it	It often gets it wrong	It learned your hands as part of banana
Use a yellow pencil case	It may shout banana	It learned yellow, not fruit
Record 200 of one class, 20 of another	It favours the big class	Unbalanced data quietly skews the model
Delete the nothing class and retrain	An empty desk becomes apple	A model must answer with what it has

Record the failures

A class list of ways to fool a model is a better record of this lesson than any screenshot of it working.

LEVEL UP

Finished early?

Go further

1 

Build a **four-class** model: ripe, unripe, bruised, nothing

Much harder. The classes overlap, and the bars show it.

2 

Export the model and run it in a **web page**

Export Model, TensorFlow.js, and paste the snippet.

3 

Test one pair's model on **another pair's** fruit

Accuracy usually drops. Ask out loud why.

4 

Try **Machine Learning for Kids** and compare

Same idea, different tool, text instead of pictures.

Check yourself

Can I really do it? Prove it.

1 Collect and **label** training examples

Proof: three classes, roughly 40 varied samples each

2 **Train** a model and test it on new pictures

Proof: it named fruit it had never seen before

3 Explain why **varied** examples beat more of the same

Proof: I named three things that fooled my model and why

4 Measure **accuracy** on held-back data

Proof: I have a before and an after number on the same 12 photos

Rate yourself: red = not yet · amber = nearly · green = yes, and I can show someone

Show what you know

What did we learn?

Five things to keep

- A model learns from labelled examples, not rules.
- A photo reaches it as a grid of numbers.
- Varied examples beat more of the same.
- Test on pictures it has never seen, or you learn nothing.
- A model only knows what it was shown.

Exit ticket

Your model calls a yellow pencil case a banana. You may not change the code. What exactly would you do?

Next class

AI-03 Nearest Neighbours: the simplest learning method there is, and you can do it on paper.